

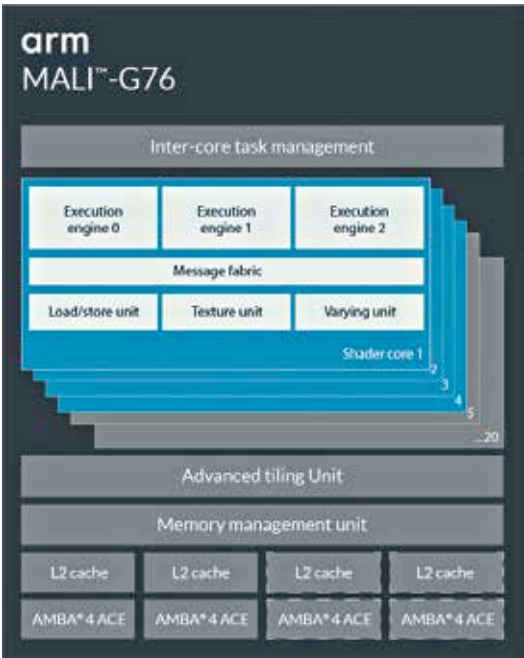
sides, which can access other components through vents bored through the substrate. In terms of component density, the latest iPhones have more tightly packed components than even the Apple Watch. There are four GPU cores, a discrete video subsystem, a neural processing unit, a display engine and a depth engine. The NPU and the depth engine work together to provide the bokeh effect of the camera, effectively separating the subject from the background.

Comparing the performance of the SoCs is not an easy task, and as they say, the devil is in the details. Benchmarks are not really an indicator of how well the phone can perform. Most Android flagships use very similar CPU and GPU configurations, even if they are on different SoCs. Apple benefits from the advantage of having an intimate relationship between its in house patented hardware and in house proprietary software, both of which it develops together. This is why Apple or Google hardware can prove themselves with half the RAM against flagships that are half the cost. It is not the bigger number on the specifications, but how well these

various pieces of hardware are managed, collectively and individually, that make a difference. If looking at benchmarks, it is important to check performance charts of multiple runs, to check how the device performs over time. Usually, devices cannot manage sustained performances, and with repeated runs, most devices typically score less and less. There are just too many factors that determine the end user experience, and this depends on all the other components of a phone, how they are put together, and even down to the choices made in the development of individual applications. Any kind of comparison of performance, essentially is always a tiny fraction of the whole picture.

THE CORES

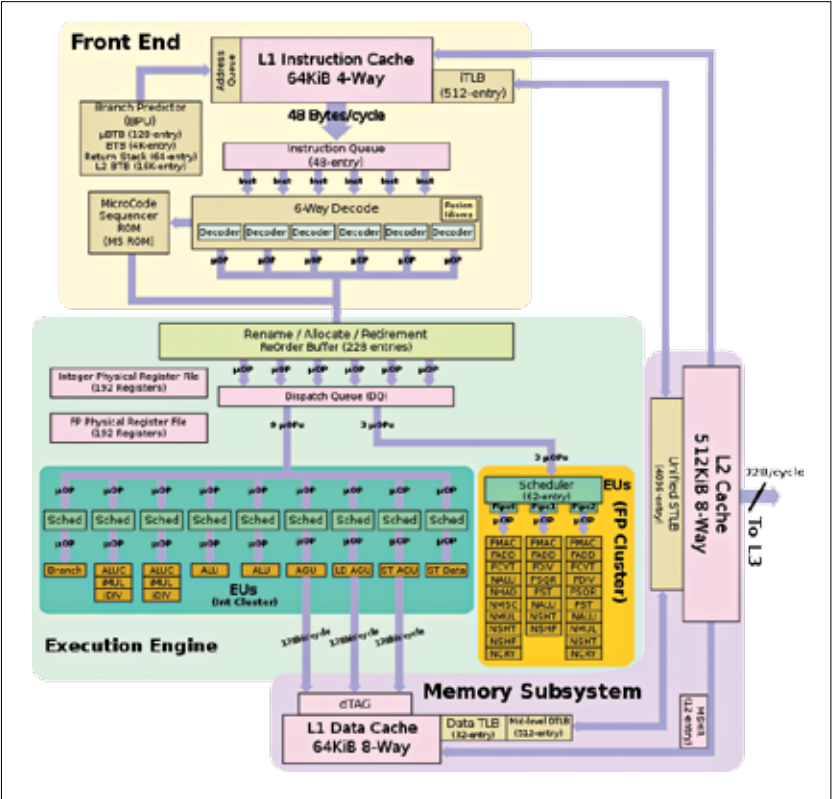
The core, also known as the micro-architecture, is again made by a select few companies, and have specific



The Mali G76 block diagram

Credit: ARM

implementations in different SoCs. The same SoC manufacturer can use the same cores differently in two different generations of the SoC. The Kirin 980 uses Cortex A76 and Cortex A75 cores. The Exynos 9820 uses a triple CPU setup, and uses Mongoose 4, Cortex A75 and Cortex A55 cores. The Helio P90 also uses two Cortex A75 and six Cortex A55 cores. The Snapdragon 855 uses eight Kryo 485 cores. As mentioned above, these are divided into four Kryo 485 Gold cores, and four Kryo 485 Silver cores. The Kryo 485 Gold cores are derived from Cortex A76 cores, while the Kryo 485 Silver cores are derived from the Cortex A55. Additionally, one of the Kryo 485 Gold cores has a higher clock speed (2.84GHz, 512KB) and cache compared to the others (2.42GHz, 256KB). Finally, the A12 uses two high performance cores called Vortex and four energy efficient cores called Tempest. The important thing to understand here is that the SoC is much more than the sum of its parts, and there are many design decisions that are not easily represented on a box packaging. The Cortex A55 cores in the Helio P90 is clocked at 2.0GHz, while they are clocked at 1.80GHz in the Kirin 980. ARM claims that the Cortex A76 is specifically designed for sustained high performance applica-



The Mongoose 4 Block Diagram